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In [ ]: ASSIGNMENT NO : 3
        Use Naïve Bayes or TF-IDF + SVM to classify emails as phishing or legitimate b
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In [1]: # PHISHING EMAIL CLASSIFICATION
        # TF-IDF + NAIVE BAYES
        import pandas as pd
        from sklearn.model_selection import train_test_split
        from sklearn.feature_extraction.text import TfidfVectorizer
        from sklearn.naive_bayes import MultinomialNB
        from sklearn.metrics import accuracy_score, classification_report, confusion_m
```

```
In [2]: data = {
        "text": [
            "Your college timetable has been updated. Please check the student por",
            "Your electricity bill for this month is now available in your account",
            "Your order has been shipped and will arrive tomorrow.",
            "The meeting is scheduled for Monday at 10 AM. Please join on time.",
            "Your bank statement for this month is available in the official banki",

            "Urgent! Your account will be blocked today. Click the link to verify ",
            "Congratulations! You have won a cash prize. Click here to claim your",
            "Your bank account has been suspended. Enter your password and OTP to",
            "Your payment failed. Click this link and enter your card details to r",
            "You have received a special offer. Confirm your account password now"
        ],
        "label": [
            0, 0, 0, 0, 0,
            1, 1, 1, 1, 1
        ]
    }
```

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In [3]: df = pd.DataFrame(data)
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In [4]: df
```

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Out[4]:
```

	text	label
0	Your college timetable has been updated. Pleas...	0
1	Your electricity bill for this month is now av...	0
2	Your order has been shipped and will arrive to...	0
3	The meeting is scheduled for Monday at 10 AM. ...	0
4	Your bank statement for this month is availabl...	0
5	Urgent! Your account will be blocked today. Cl...	1
6	Congratulations! You have won a cash prize. Cl...	1
7	Your bank account has been suspended. Enter yo...	1
8	Your payment failed. Click this link and enter...	1
9	You have received a special offer. Confirm you...	1

```
In [ ]: # 0 = Legitimate and 1 = Phishing
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In [5]: df.head()
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Out[5]:
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	text	label
0	Your college timetable has been updated. Pleas...	0
1	Your electricity bill for this month is now av...	0
2	Your order has been shipped and will arrive to...	0
3	The meeting is scheduled for Monday at 10 AM. ...	0
4	Your bank statement for this month is availabl...	0

```
In [6]: df.tail()
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```
Out[6]:
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	text	label
5	Urgent! Your account will be blocked today. Cl...	1
6	Congratulations! You have won a cash prize. Cl...	1
7	Your bank account has been suspended. Enter yo...	1
8	Your payment failed. Click this link and enter...	1
9	You have received a special offer. Confirm you...	1

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In [7]: df.isnull().sum()
```

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Out[7]: text      0
label      0
dtype: int64
```

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In [8]: df.columns
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Out[8]: Index(['text', 'label'], dtype='object')
```

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In [9]: X = df["text"]
y = df["label"]
```

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In [10]: X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.20,
                                                    random_state=42)
```

```
In [11]: tfidf = TfidfVectorizer(lowercase=True,stop_words="english")

X_train_tfidf = tfidf.fit_transform(X_train)
X_test_tfidf = tfidf.transform(X_test)
```

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In [12]: model = MultinomialNB()
model.fit(X_train_tfidf, y_train)
```

```
Out[12]: MultinomialNB()
```

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In [15]: y_pred = model.predict(X_test_tfidf)

print("Predicted:", y_pred)
print("Actual:", y_test.values)
```

```
Predicted: [1 0]
Actual: [1 0]
```

```
In [14]: accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
print("Accuracy (%)":, accuracy * 100)
```

```
Accuracy: 1.0
Accuracy (%): 100.0
```

```
In [16]: print("Classification Report:")
print(classification_report(y_test,y_pred,labels=[0, 1],
                           target_names=["Legitimate", "Phishing"],
                           zero_division=0))
```

```
Classification Report:
              precision    recall  f1-score   support

 Legitimate      1.00      1.00      1.00         1
  Phishing      1.00      1.00      1.00         1

 accuracy              1.00         2
 macro avg      1.00      1.00      1.00         2
weighted avg      1.00      1.00      1.00         2
```

```
In [17]: print("Confusion Matrix:")
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
Confusion Matrix:
[[1 0]
 [0 1]]
```

```
In [18]: new_email = [
    "Urgent! Your account has been suspended.Click the link to verify your pas

new_email_tfidf = tfidf.transform(new_email)

prediction = model.predict(new_email_tfidf)
```

```
if prediction[0] == 1:  
    print("Prediction: PHISHING")  
else:  
    print("Prediction: LEGITIMATE")
```

Prediction: PHISHING